WORKED SOLUTION

STEP II 2022, Question 4

Pure • Modulus Functions

STEP II 2022, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

A modulus sign only matters at the point where its contents change sign, so on each interval between such points every modulus can be replaced by plus or minus its contents. Going the other way, a term $A|x-p|$ puts a corner at $x=p$ where the gradient jumps by $2A$. So to build a single formula, we match the gradient jumps (and, in the last part, the jumps in the x^2 coefficient) and then fix the constant.

Solution

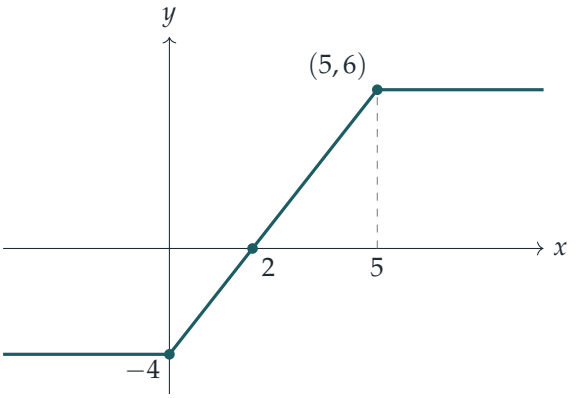
Part (i)

The critical points are $x=0$ (where $|x|$ switches) and $x=5$ (where $|x-5|$ switches).

- For $x \leq 0$: $|x| = -x$ and, since $x-5<0$, $|x-5| = 5-x$. So $\mathrm{f}(x) = -x-(5-x)+1 = -4$.
- For $0 \leq x \leq 5$: $|x| = x$ and $|x-5| = 5-x$. So $\mathrm{f}(x) = x-(5-x)+1 = 2x-4$.
- For $x \geq 5$: $|x| = x$ and $|x-5| = x-5$. So $\mathrm{f}(x) = x-(x-5)+1 = 6$.

At $x=0$ and $x=5$ both relevant expressions are valid (each modulus is zero there or has a definite sign), so the overlapping endpoints are consistent: $-4 = 2(0)-4$ and $2(5)-4 = 6$. This is the required piecewise form.

The graph is horizontal at height -4, rises with gradient 2 from $(0,-4)$ through $(2,0)$ to $(5,6)$, then is horizontal at height 6.



Part (ii)

Part (i) suggests trying

$$g(x) = A|x| + B|x-5| + Cx + D.$$

Passing from left to right through $x = 0$, the term $A|x|$ changes from $-Ax$ to Ax, so the gradient jumps by $2A$; similarly $B|x-5|$ makes the gradient jump by $2B$ at $x = 5$. The given g has gradients -1, 3, 1 on the three pieces, so

$$2A = 3 - (-1) = 4, \qquad 2B = 1 - 3 = -2,$$

giving $A = 2$, $B = -1$. For $x \geq 5$ the formula is $2x - (x-5) + Cx + D = (1+C)x + 5 + D$, and this must be $x + 10$, so $C = 0$ and $D = 5$. Hence

$$\boxed{g(x) = 2|x| - |x-5| + 5.}$$

Checking the other two pieces: for $x \leq 0$, $-2x - (5-x) + 5 = -x$; for $0 \leq x \leq 5$, $2x - (5-x) + 5 = 3x$. Both agree with the definition of g.

? **Ask yourself** Why did a formula of this shape have any chance of working?

Each piece of g is linear and the pieces join up continuously (0 at $x = 0$ and 15 at $x = 5$ from both sides). A sum of moduli of linear terms plus a linear function is always continuous and piecewise linear, with corners only where the moduli switch. If g had a jump, no such formula could produce it.

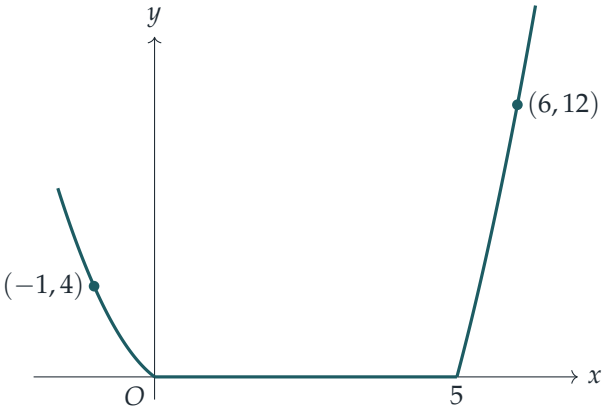
Part (iii)

Now the switching points are $x = 0$ (for $|x|$) and the roots of $x(x-5)$, which are again 0 and 5. The product $x(x-5)$ is positive for $x < 0$ and $x > 5$, and negative between.

- For $x \leq 0$: $h(x) = x^2 - x + 4x + (x^2 - 5x) = 2x^2 - 2x = 2x(x-1)$.
- For $0 \leq x \leq 5$: $h(x) = x^2 - x - 4x - (x^2 - 5x) = 0$.

- For $x \geq 5$: $\mathrm{h}(x) = x^2 - x - 4x + (x^2 - 5x) = 2x^2 - 10x = 2x(x-5)$.

So h is identically zero on $[0,5]$. To the left it is part of the parabola $y = 2x(x-1)$: it meets the origin with gradient $4x - 2 = -2$, and it increases without bound as $x \to -\infty$ (for example $\mathrm{h}(-1) = 4$). To the right it is part of $y = 2x(x-5)$: it leaves $(5,0)$ with gradient $4x - 10 = 10$ (for example $\mathrm{h}(6) = 12$). The two arms are *not* mirror images: the right arm leaves the axis five times as steeply. The graph is continuous, with corners at $(0,0)$ and $(5,0)$, and $\mathrm{h}(x) \geq 0$ everywhere.



Part (iv)

Part (iii) shows the tool we need: $|x(x-5)|$ equals $x^2 - 5x$ outside $[0,5]$ and $-(x^2-5x)$ inside, so it can switch an x^2 term on and off. The function k has x^2 coefficient 0 outside $[0,5]$ and 2 inside. In $ax^2 + d|x(x-5)|$ the coefficient of x^2 is $a+d$ outside and $a-d$ inside, so we need $a+d=0$ and $a-d=2$, that is, $a=1$ and $d=-1$. Thus $x^2 - |x(x-5)|$ has the right quadratic behaviour, and

$$x^2 - |x(x-5)| = \begin{cases} 5x & x \leq 0 \text{ or } x \geq 5, \\ 2x^2 - 5x & 0 \leq x \leq 5. \end{cases}$$

Subtracting this from k leaves a piecewise linear remainder:

$$\mathrm{k}(x) - \left(x^2 - |x(x-5)|\right) = \begin{cases} 5x & x \leq 5, \\ 50 - 5x & x \geq 5. \end{cases}$$

This has a single corner at $x = 5$, where the gradient jumps from 5 to -5, a jump of -10; so it is $-5|x-5| + (\text{linear})$. For $x \leq 5$, $-5|x-5| = 5x - 25$, so we add 25 to get $5x$. This suggests

$$\boxed{\mathrm{k}(x) = x^2 - |x(x-5)| - 5|x-5| + 25.}$$

Verification. We substitute on each interval.

- For $x \leq 0$: $x(x-5) \geq 0$ and $x - 5 < 0$, so the formula gives $x^2 - (x^2 - 5x) - 5(5 - x) + 25 = 5x + 5x = 10x$.

- For $0 \le x \le 5$: $x(x-5) \le 0$ and $x-5 \le 0$, so it gives $x^2 + (x^2 - 5x) - 5(5-x) + 25 = 2x^2$.
- For $x \ge 5$: $x(x-5) \ge 0$ and $x-5 \ge 0$, so it gives $x^2 - (x^2 - 5x) - 5(x-5) + 25 = 50$.

These are exactly the three pieces of k, so the formula is correct for every real x.

The single formula isn't unique. For example, $|x(x-5)| = |x|\,|x-5|$, so the answer can be rewritten, and other combinations (such as ones using $x|x|$) also work. Any answer is fine if it passes the three-interval check above.

Checks

- Spot values: $\mathrm{g}(-2) = 2$ from both the definition and $2(2) - 7 + 5$; $\mathrm{g}(7) = 17 = 14 - 2 + 5$. For k: $\mathrm{k}(3) = 18$ and $9 - |-6| - 10 + 25 = 18$; $\mathrm{k}(-1) = -10$ and $1 - 6 - 30 + 25 = -10$.
- Continuity is a quick sanity test: every single formula built from moduli is continuous, and indeed g and k agree at the joins ($\mathrm{k}(5) = 50 = 2 \cdot 5^2$).
- All four formulas were compared with their piecewise forms by computer at 10^5 random points in $[-20, 25]$, with agreement to rounding error.

Common slips

- Sketching h with symmetric arms. The left arm is $2x(x-1)$ and the right arm is $2x(x-5)$, with gradients -2 and 10 at the corners.
- Forgetting that $|x(x-5)|$ changes sign at *two* points, so in (iii) it flips in the middle interval, not just at 0.
- Writing down a formula for k without substituting into all three intervals; the verification is required.

Beyond STEP

The gradient-jump bookkeeping here is a first taste of a general fact: every continuous piecewise linear function on $\mathbb{R}$ with finitely many corners is a linear function plus a combination of terms $|x-p|$. This is the idea behind linear splines in numerical analysis and the "ReLU" units of neural networks, since $\max(x, 0) = \frac{1}{2}(x + |x|)$. For a careful treatment of continuity and functions defined piece by piece, see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.